

22) Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = mx + b$.
 a. Show that f is a linear transformation when $b = 0$

Soln: $f(x) = mx + 0 = mx$ (Algebraic property iii)
 $f(x+y) = m(x+y) = mx + my$ (Algebraic property v) $= f(x) + f(y)$
 $\therefore$ Condition 1 satisfied.
 $f(cx) = m(cx) = (mc)x$ (Algebraic property vii) $= (cm)x = c(mx)$ (Algebraic property vii) $= c \cdot f(x)$
 $\therefore$ Condition 2 satisfied. By Defn, f is a linear transformation when $b = 0$.
 $\square$

b. Find a property of a linear transformation that is violated when $b \neq 0$.

Soln: $f(x+y) = \cancel{m(x+y)+b} = mx + (my+b)$ (Algebraic property ii)
 $f(x) = mx+b$ $f(y) = my+b$ $f(x)+f(y) = (mx+b) + (my+b)$
 $\therefore f(x+y) \neq f(x) + f(y)$. The condition 1 of the definition of linear transformation is violated.

c. Why is f called a linear function?

Ans: Because f is linear on the unknown variable x . $\square$
 30) An affine transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ has the form $T(x) = Ax + b$ with A an $m \times n$ matrix and b in $\mathbb{R}^m$. Show that T is not a linear transformation when $b \neq 0$.

Solution: $T(x+y) = A(x+y) + b = (Ax + Ay) + b$ (Theorem 5(a)) $= Ax + (Ay + b)$
 $T(x) = Ax + b$ $T(y) = Ay + b$ $\therefore T(x) + T(y) = (Ax + b) + (Ay + b) = Ax + (b + (Ay + b))$
 $= Ax + (Ay + (b + b))$ (Algebraic property ii) $= Ax + (Ay + (b + b))$ (Algebraic property i)
 $= Ax + (Ay + (b + b))$ (Algebraic property ii) $= (Ax + Ay) + (b + b)$ (Algebraic property ii)

Let $T(x+y) = T(x) + T(y)$. $\therefore (Ax + Ay) + b = (Ax + b) + (Ay + b)$
 $\Rightarrow -(Ax + Ay) + (Ax + Ay) + b = -(Ax + Ay) + (Ax + b) + (Ay + b)$ (Algebraic property ii)
 $\Rightarrow -(Ax + Ay) + (Ax + Ay) + b = -(Ax + Ay) + (Ax + Ay) + (b + b)$